

MIMIC-III Processed Data — Files, Shapes, and Relationships

Companion to `MIMIC Data/MIMIC-III Data Instruction.pdf`. The PDF describes **what** each file contains; this note adds **shapes**, **column names**, **how the files line up**, and the **subset relationships** between them.

All paths below are relative to `MIMIC Data/`.

1. Files in `processed-data/`

File	Shape	Header?	Index/key
<code>input.csv</code>	35,643 × 26	no	row-aligned with <code>valid_aids.csv</code>
<code>valid_aids.csv</code>	35,643 × 1	no	the column is <code>HADM_ID</code>
<code>label.csv</code>	35,304 × 3	yes (<code>HADM_ID</code> , <code>sensor</code> , <code>time</code>)	<code>HADM_ID</code>
<code>imputed-normed-ep_1_24.npz</code>	dict of arrays, leading dim = 35,643	—	row-aligned with <code>valid_aids.csv</code>

Column names for `input.csv` (V1..V26 from the PDF)

V1	age	V14	potassium_min
V2	heartrate_max	V15	potassium_max
V3	heartrate_min	V16	sodium_min
V4	sysbp_max	V17	sodium_max
V5	sysbp_min	V18	bicarbonate_min
V6	tempc_max	V19	bicarbonate_max
V7	tempc_min	V20	bilirubin_min
V8	PaO2FiO2_vent_min	V21	bilirubin_max
V9	urineoutput	V22	gcs_min
V10	bun_min	V23	AIDS (categorical 0/1)
V11	bun_max	V24	HEM (categorical 0/1)
V12	wbc_min	V25	METS (categorical 0/1)
V13	wbc_max	V26	Adm_type (categorical 0/1/2)

The PDF says "17 features", which expand into **26 summary statistics** (min/max pairs for most continuous features).

Keys in `imputed-normed-ep_1_24.npz`

<code>ep_tdata</code>	(35643, 24, 15)	time-varying features (15 vars × 24 hrs)
<code>ep_tdata_masking</code>	(35643, 24, 15)	missingness mask for <code>ep_tdata</code>
<code>ep_data</code>	(35643, 360)	<code>ep_tdata</code> flattened (24*15=360)
<code>ep_data_masking</code>	(35643, 360)	missingness mask for <code>ep_data</code>
<code>adm_features_all</code>	(35643, 5)	time-static admission features
<code>adm_labels_all</code>	(35643, 6)	admission-level labels
<code>y_mor</code>	(35643, 1)	in-hospital mortality
<code>y_icd9</code>	(35643, 20)	ICD-9 chapter labels
<code>y_los</code>	(35643,)	length of stay

The leading dimension (35,643) of every array is **row-aligned with `valid_aids.csv`** in the same order.

2. Files in `raw-data/`

File	Notes
<code>ADMISSIONS.csv</code>	source for length of stay = <code>DISCHTIME</code> - <code>ADMITTIME</code> ; can be merged on <code>HADM_ID</code>
<code>PATIENTS.csv</code>	demographics; merge via <code>SUBJECT_ID</code> (chase through <code>ADMISSIONS</code> first)

3. Files in `imputed-split-data/`

10 random 3:1:1 train/valid/test splits, mean-imputed and standardized using train-fold means/stds.

<code>imputed-split-data/series/</code>	<code>{train,valid,test}_{1..10}.pkl</code>	time-series tensors
<code>imputed-split-data/static/</code>	<code>{train,valid,test}_{1..10}.pkl</code>	static features

4. Subset relationships

The `HADM_ID` (hospital admission ID) is the join key across all files.

```
HADM_ID(input.csv) == HADM_ID(valid_aids.csv) == leading axis of npz
                        ^
                        |
                        HADM_ID(label.csv)
```

Concretely, with the IDs computed in `merge_data.py` :

| Set | |HADM_ID| | Notes | |---|---|---| | `valid_aids.csv` (= `input.csv` 's rows) | **35,643** | row-aligned to `input.csv` and to the `npz` | | `label.csv` | **35,304** | strict subset of the above | | `valid_aids.csv` $\cap$ `label.csv` | **35,304** | every label row matches an input row | | `valid_aids.csv` $\setminus$ `label.csv` | **339** | input rows with no label (NaN after left join) | | `label.csv` $\setminus$ `valid_aids.csv` | **0** | nothing label-only |

So `label.csv` $\subset$ `input.csv` (by `HADM_ID`), and **339** of the 35,643 admissions in `input.csv` have no survival outcome.

5. Merge artifacts produced by `merge_data.py`

Outputs land in `mimic_analysis/merged_data/` :

File	Shape	What it is
<code>input_named.csv</code>	35,643 × 27	<code>HADM_ID</code> + V1..V26 (named columns)
<code>merged.csv</code>	35,643 × 29	<code>input_named</code> $\bowtie$ <code>label.csv</code> on <code>HADM_ID</code> (left join, so all 35,643 rows kept; 339 have NaN <code>ensor / time</code>)

Why `merged.csv` has more rows than `label.csv`

```
35,643 (merged.csv)
= 35,304 (HADM_IDs in both input and label, with censor/time)
+ 339 (HADM_IDs in input only, censor/time = NaN)
```

If you only want labeled admissions, swap the left join in `merge_data.py` for an inner join — that produces a **35,304 × 29** frame.

Event / censoring breakdown (from `merged.csv`)

Among the **35,304** rows that have a label:

censor	Count	Meaning
1	13,765	event observed (death)
0	21,539	censored (alive at end of follow-up)

`time` is the time to mortality after admission (units per the original benchmark — typically years).

6. The 339 unlabeled admissions

`diagnose_unlabeled.py` cross-references the 339 input-only `HADM_ID` s with `ADMISSIONS.csv` and `PATIENTS.csv` . Findings:

Cohort characteristics (unlabeled vs labeled)

Statistic	Unlabeled (n=339)	Labeled (n=35,304)
Mean age at admission	94.4 yr	74.4 yr
Median LOS	1.2 days	7.1 days
Died in hospital (<code>HOSPITAL_EXPIRE_FLAG=1</code>)	98.5%	10.0%
Has <code>DEATHTIME</code> recorded	98.5%	10.0%
Has <code>DOD</code> (date of death) on file	100%	36.4%
Emergency admission	95.9%	84.0%

The unlabeled cohort is a tightly-clustered group: very old patients, very short stays, almost all dying in hospital. **Every single one has a recorded date of death** in `PATIENTS.csv` .

One admission per patient

```
339 unlabeled admissions ↔ 339 unique SUBJECT_IDs
```

None of these subjects has any *other* admission in `valid_aids.csv` . They are all single-admission patients.

Model interpretation: these are death-censored subjects

In the recurrent-event readmission model used here, the **event of interest is hospital readmission**, not death. Death precludes any future readmission, so for these 339 patients death acts as a **censoring mechanism** — specifically a **death censor** — not as an observed event.

This explains the upstream label-generation choice cleanly:

- These patients died in their first (and only) admission.
- They will never have a subsequent readmission to observe.
- Their contribution to the recurrent-event likelihood is purely the "no event observed by time T" censoring term, with $T = DOD - ADMITTIME$.

So the 339 missing `label.csv` rows are not noise or missingness — they correspond to a structurally well-defined subgroup that the readmission model treats as right-censored at time of death.

Implications for downstream use

If you want to use all 35,643 admissions:

Field	Value for the 339
<code>censor</code>	0 (right-censored — no readmission event observed)
<code>time</code>	$DOD - ADMITTIME$ (or $DISCHTIME - ADMITTIME$ if treating discharge as the censoring time)

Computing this is straightforward because `DOD` is known for all 339. The diagnostic CSV (`merged_data/unlabeled_diagnostic.csv`) carries the fields needed to perform this imputation.

If you only want patients at-risk for readmission, **drop the 339** and work with the 35,304 labeled rows — that is what `label.csv` already encodes.

7. Reproducing this layout

```
python3 mimic_analysis/merge_data.py
```

Reads from `MIMIC Data/processed-data/` and writes `mimic_analysis/merged_data/{input_named.csv, merged.csv}`. Re-run any time `processed-data/` changes upstream.